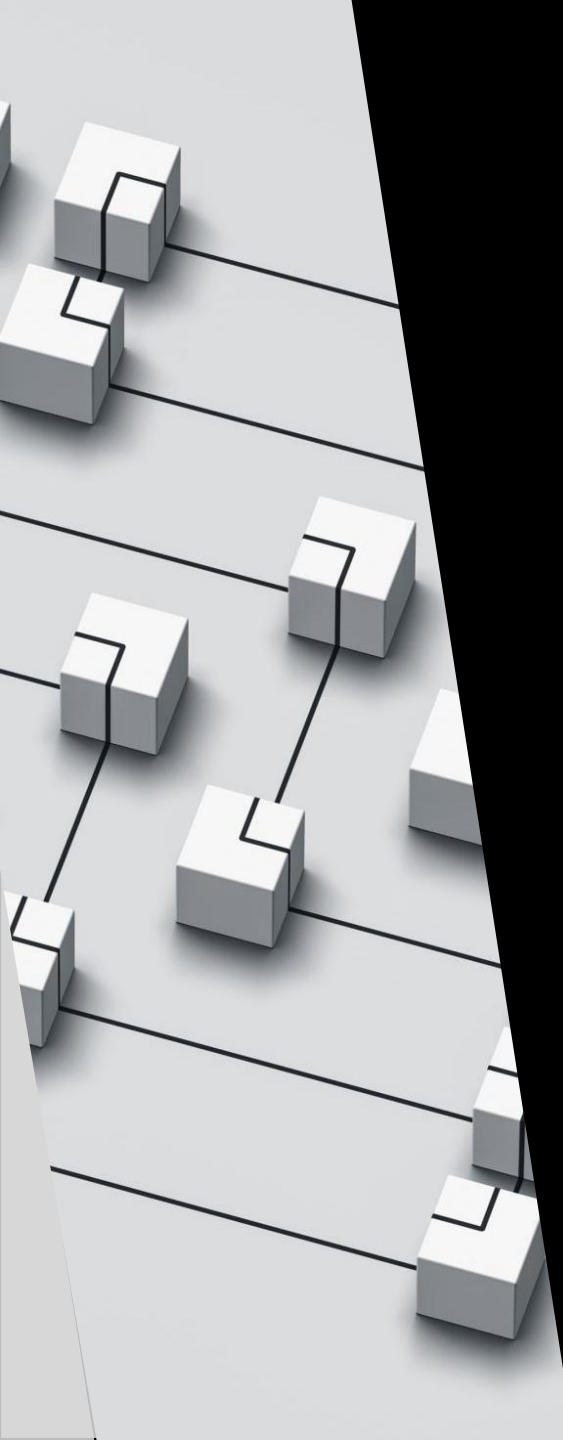


Hybrid Telco Churn and Geospatial Risk Analysis

(Integrating Infrastructure Health into Predictive Customer Retention)



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April 2026



Executive Summary

- ▶ **The Problem:** Traditional churn models overlook physical infrastructure failures as primary drivers of attrition.
- ▶ **The Innovation:** A hybrid model integrating customer behavior with engineered network indicators (Signal Stability & Outage History).
- ▶ **The Strategy:** Optimized a **Logistic Regression** model to prioritize coverage, achieving an **84% AUC** and **85% churn capture (Recall)** via threshold tuning.
- ▶ **Key Discovery:** Identified a **0.6 signal stability threshold** as the critical technical breaking point for churn, especially among high-value Fiber Optic users.

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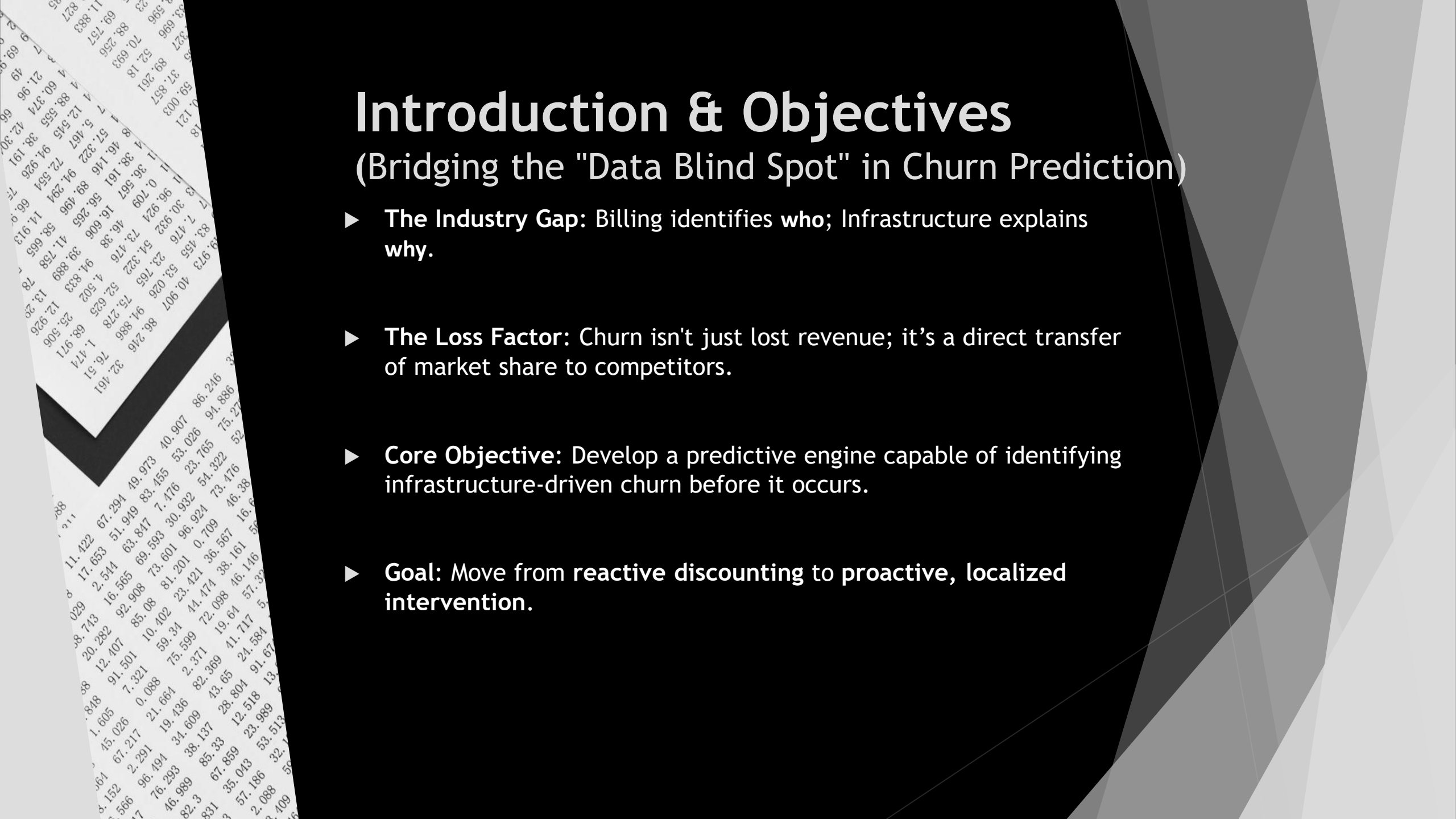
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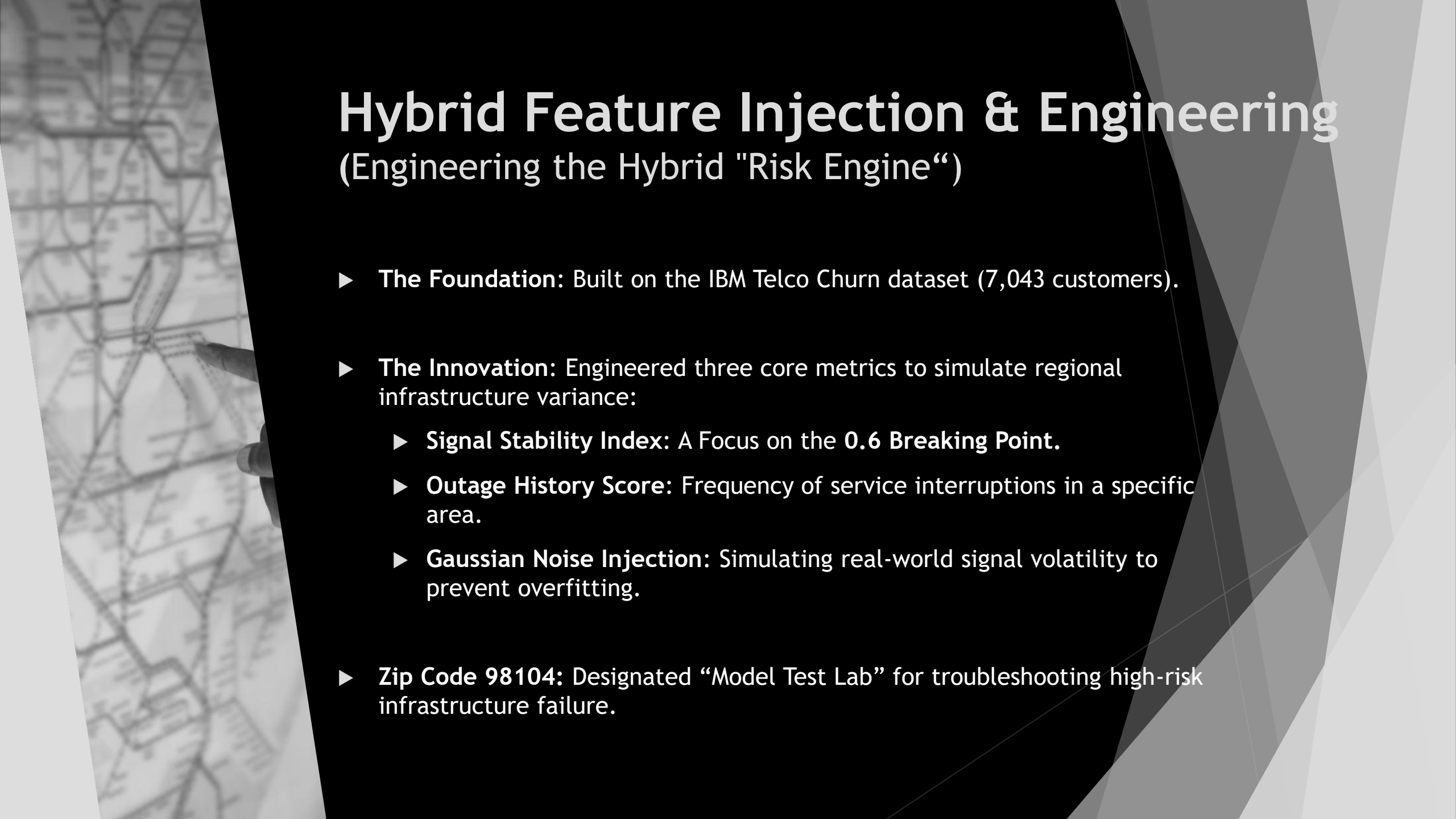
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Introduction & Objectives

(Bridging the "Data Blind Spot" in Churn Prediction)

- ▶ **The Industry Gap:** Billing identifies who; Infrastructure explains why.
- ▶ **The Loss Factor:** Churn isn't just lost revenue; it's a direct transfer of market share to competitors.
- ▶ **Core Objective:** Develop a predictive engine capable of identifying infrastructure-driven churn before it occurs.
- ▶ **Goal:** Move from reactive discounting to proactive, localized intervention.



Hybrid Feature Injection & Engineering

(Engineering the Hybrid "Risk Engine")

- ▶ **The Foundation:** Built on the IBM Telco Churn dataset (7,043 customers).
- ▶ **The Innovation:** Engineered three core metrics to simulate regional infrastructure variance:
 - ▶ **Signal Stability Index:** A Focus on the 0.6 Breaking Point.
 - ▶ **Outage History Score:** Frequency of service interruptions in a specific area.
 - ▶ **Gaussian Noise Injection:** Simulating real-world signal volatility to prevent overfitting.
- ▶ **Zip Code 98104:** Designated "Model Test Lab" for troubleshooting high-risk infrastructure failure.

Model Robustness & Optimization

Dimensionality Management:

- **Target:** Focus on infrastructure health rather than static geography.
- **Expansion:** Increased feature set from 24 to 40+ predictors via One-Hot Encoding.
- **Selection:** Replaced raw Zip Codes with dynamic, engineered geospatial metrics.
- **Outcome:** Prevented "data leakage" and forced model generalization.

Gaussian Noise Application:

- **Action:** Applied Gaussian noise to the Signal Stability Index.
- **Purpose:** Simulates real-world signal fluctuations and volatility.
- **Benefit:** Prevents overfitting on static data, ensuring real-world robustness.

Exploratory Data Analysis (EDA)

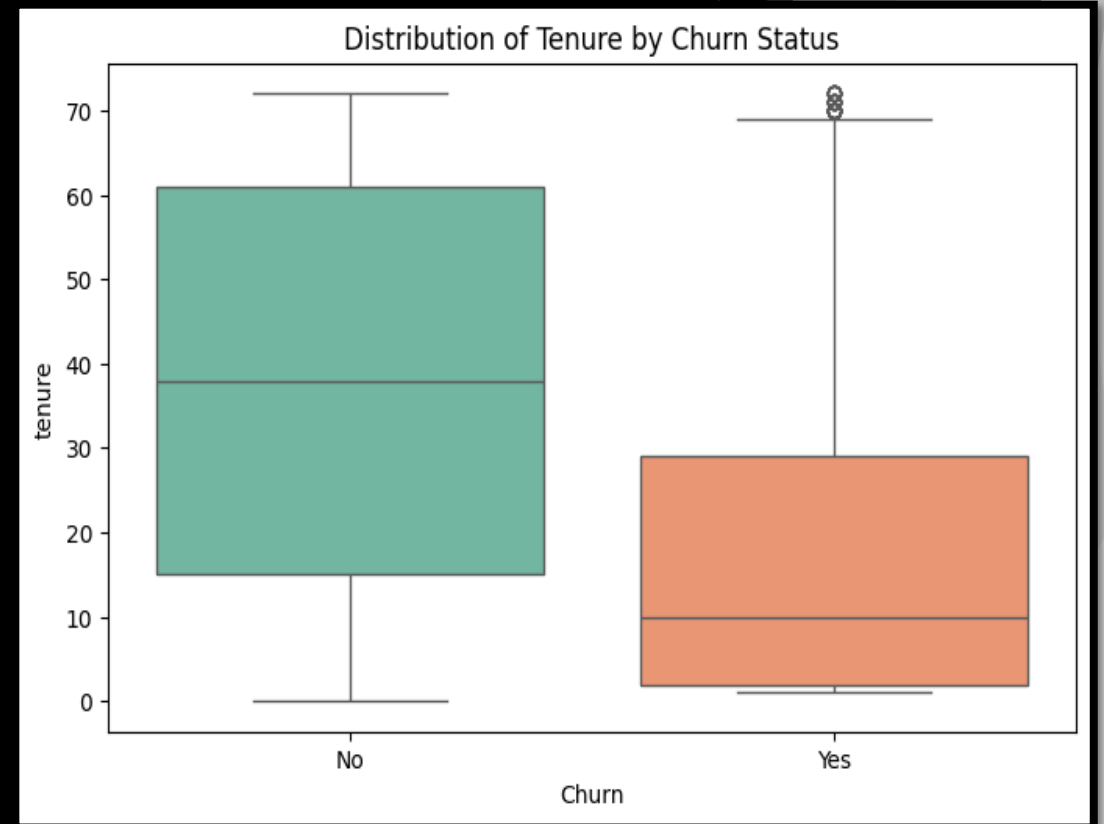
(Tenure Distribution and Churn Propensity)

Key Insight:

- ▶ **Churn Benchmark:** Median tenure for churned Customers is ~10 months vs. ~40 months for retained ones.
- ▶ **The Vulnerability Window:** 75% of churn occurs within 30 months; customers <20 months represent the highest immediate risk.
- ▶ **Loyalty Blind Spot:** Outliers show that even long-term customers churn, suggesting external triggers like infrastructure failure.

Business Impact:

- ▶ This baseline identifies our vulnerable population, but now we need to look at the 'Trouble Zones' to see what actually pushes them over the edge.



Exploratory Data Analysis (EDA)

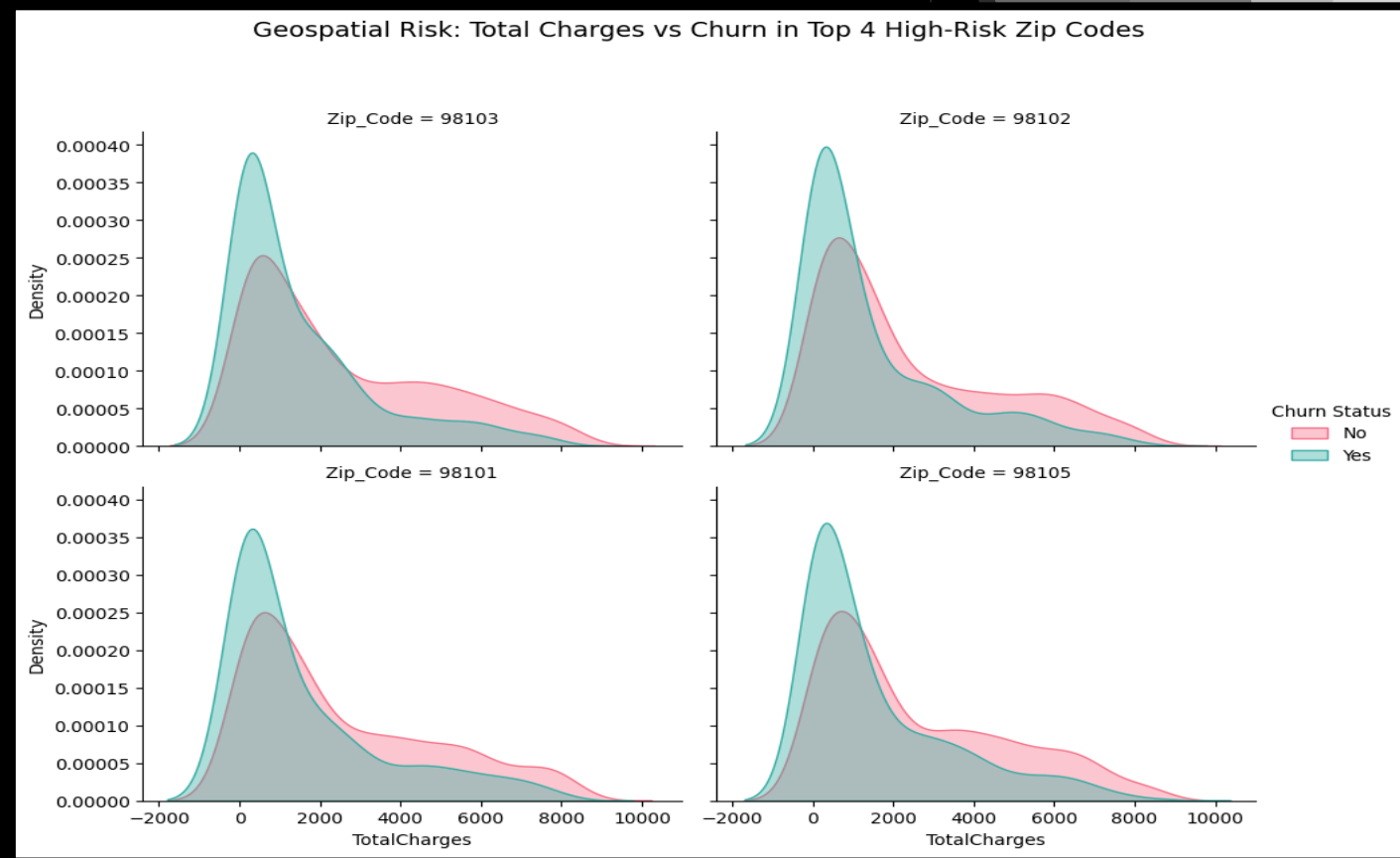
(Total Charges vs. Churn in High-Risk Zip Codes)

► Key Insight:

- **Systemic Risk:** A massive churn spike occurs under \$1,000 total charges in Zips 98101-98105.
- **The Verdict:** Regional infrastructure failure is driving immediate attrition, preventing customers from ever reaching their first year.
- **Loyalty Anchor:** Retention probability only begins to exceed churn risk after the \$2,000 spending threshold.

► Surgical Infrastructure Intervention:

- These zones is required to stop the loss of high-potential new subscribers.



Exploratory Data Analysis (EDA)

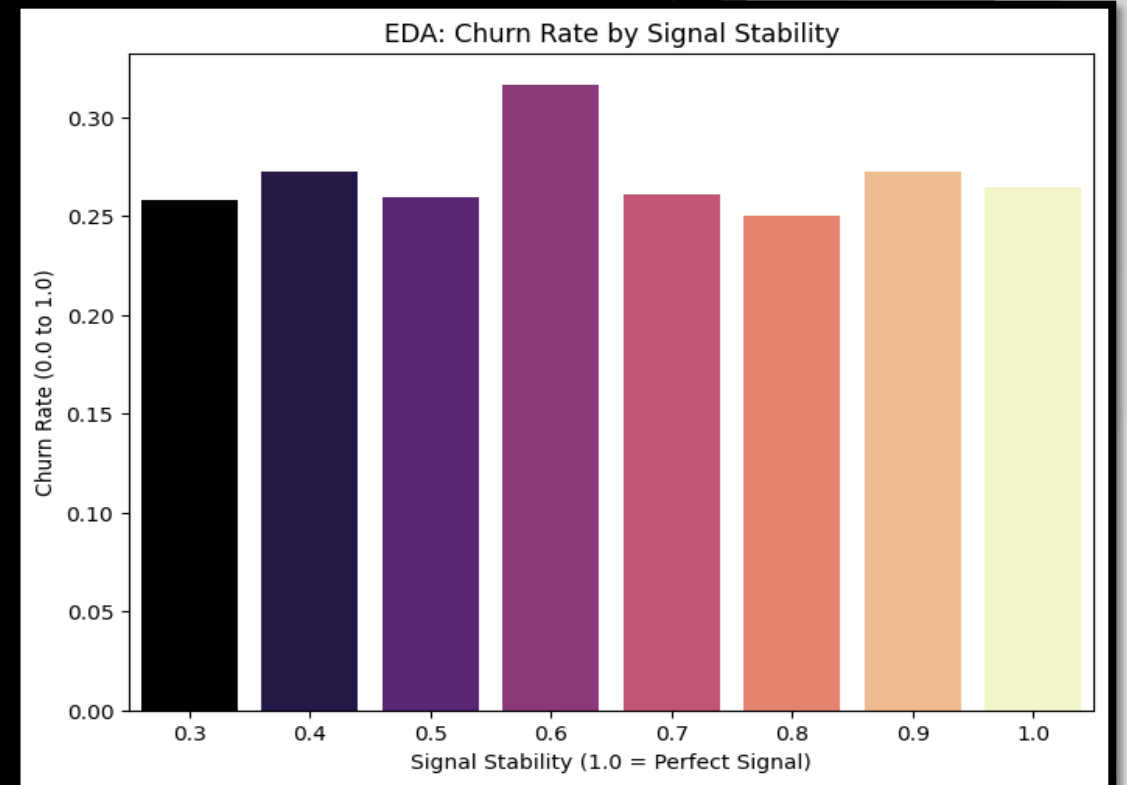
(Churn Rate Sensitivity to Signal Stability)

► Key Insight:

- **The Breaking Point:** Churn surges to 35% when Signal Stability hits 0.6.
- **Service Tolerance:** Customers tolerate fluctuations between 0.7 and 1.0; however, 0.6 represents a critical technical failure.
- **Hybrid Logic:** This confirms that once technical reliability is achieved, behavioral factors take over as the primary churn drivers.

► Strategy:

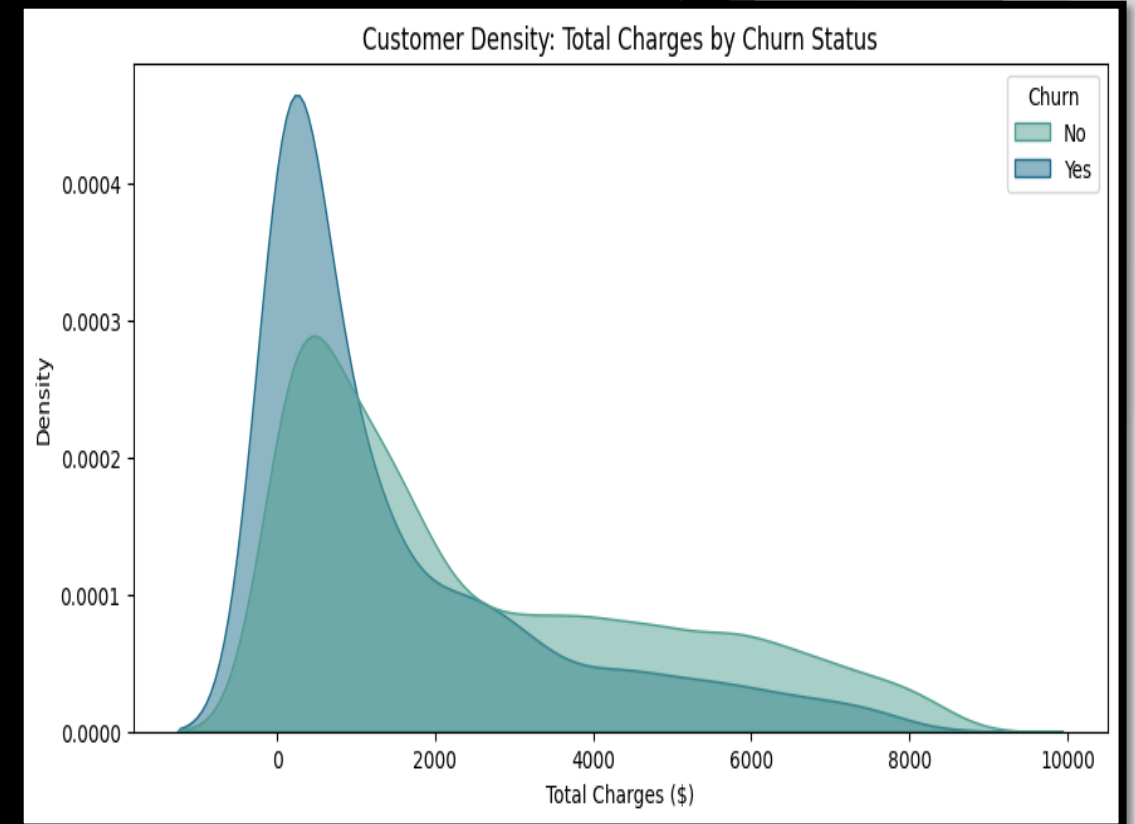
- Use 0.6 as a real-time "Geospatial Red Flag" to trigger immediate maintenance.



Exploratory Data Analysis (EDA)

(Total Charges as a Financial Risk Profile)

- ▶ **Key Insight:**
 - ▶ **Zero-Buffer (M2M):** Low-tenure/Month-to-month customers churn immediately upon signal degradation; they have no sunk cost to keep them.
 - ▶ **Retention Gap:** Total charges reflect tenure and contract depth. While Two-year contracts typically retain customers, **extreme instability (below 0.5)** still acts as a 'contract breaker.'
- ▶ **Recommendation:**
 - ▶ **Dual Strategy:** Prioritize technical repairs to stabilize the connection while simultaneously offering **contract incentives to anchor behavior.**



Exploratory Data Analysis (EDA)

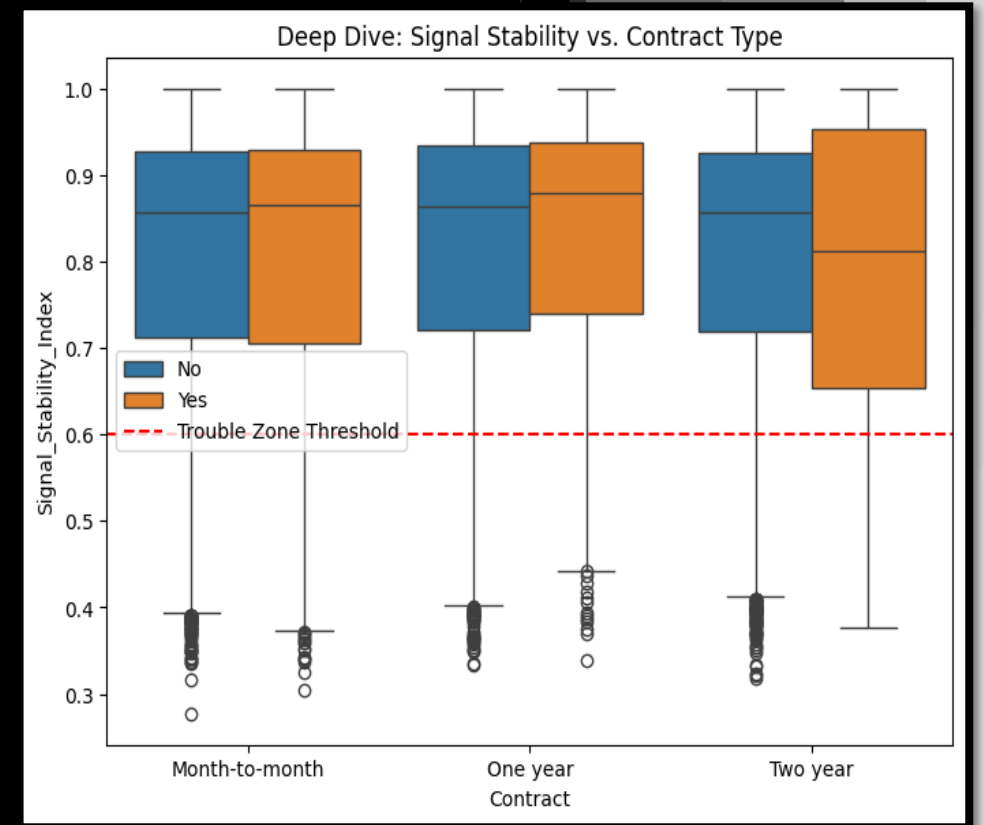
(The Technical Breaking Point of Loyalty)

► Key Insights:

- **The 0.6 Trouble Zone:** Across all contract types, the majority of customers who stay (Blue) maintain a Signal Stability Index above the 0.6 red line.
- **The "Breaker" Effect (Two-Year Contracts):** Notice the extended lower whisker for Two-Year churners; these customers were pushed to an extreme technical limit before exiting.
- **Extreme Outliers:** The presence of churned outliers (dots) as low as 0.35 proves that when technical infrastructure reaches a state of total failure, contractual loyalty disappears.

► Strategic Recommendation:

- **Proactive Retention:** Don't rely on contracts to "buffer" technical risk. Use the 0.6 threshold as a trigger for a **technical service credit** or immediate tower repair to prevent high-value, long-term losses.



Data Preprocessing & Transformation

(Preparing Data for Machine Learning)

Categorical Encoding (One-Hot):

- ▶ Converted service categories into 40+ **binary predictors**.
- ▶ Applied `drop_first=True` to eliminate redundant data (Multicollinearity).

Numerical Scaling(StandardScaler)

- ▶ **Leveling the Playing Field:** Ensures that financial magnitude does not drown out critical technical signals.

Leakage Prevention (Gold Standard):

- ▶ **Fit-on-Train Only:** Scaler was fitted exclusively on the training set to prevent "information from the future" from biasing results.
- ▶ **Authenticity:** Ensures the final 0.84 AUC is a true reflection of real-world performance.

Model Selection & Tournament

(Strategic Model Selection: Logistic Regression)

The "Tournament" Results (5-Fold CV):

- **Logistic Regression: 0.8448**
(*C=10, penalty='elasticnet', l1_ratio=1.0*)
- Gradient Boosting: 0.8470
(*n_estimators=100, lr=0.1*)
- Random Forest: 0.8322
(*n_estimators=200, max_depth=10*)

Why Logistic Regression?

Logistic Regression captures nearly the same predictive power (0.84) while offering **full transparency** for stakeholder reporting.

**Hyperparameter
Tuning: Optimized via GridSearch**

Optimization Target:

Defensive Strategy: Tuned for **85% Recall** to prioritize aggressive churn prevention in high-risk geospatial zones.

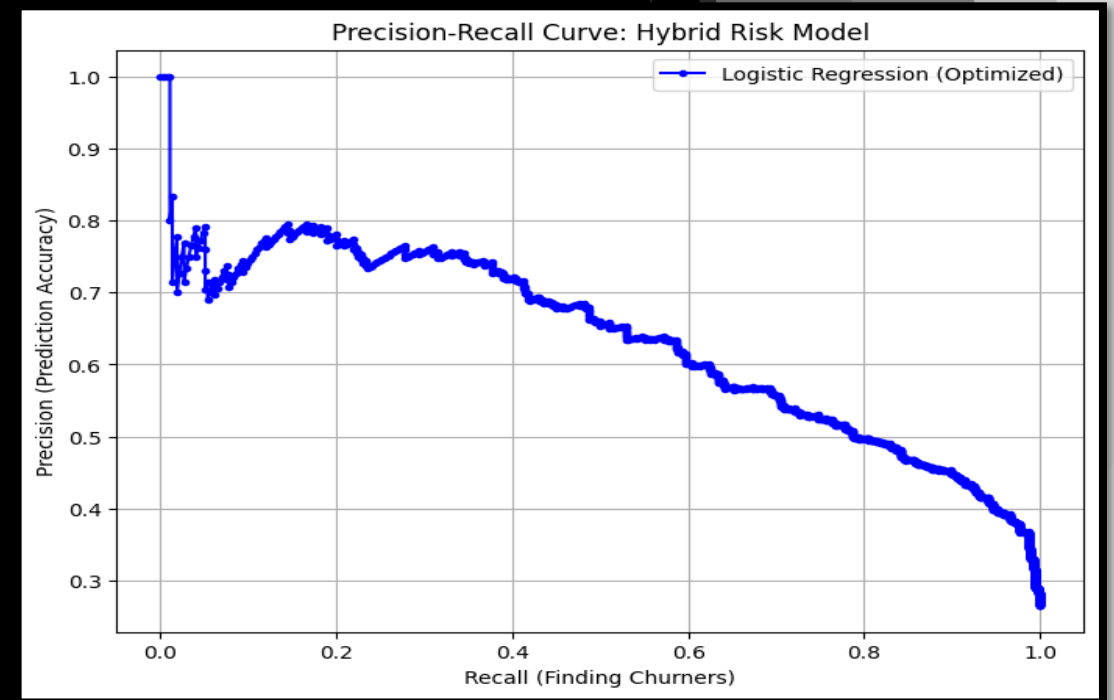
Strategic Trade-off:

Prioritizing business interpretability over a marginal **0.3% gain in raw accuracy** to ensure actionable insights.

The Defensive Strategy (PR Curve)

(The 0.43 Threshold to hit 85% Recall)

- ▶ **Strategic Balance:** Moved beyond default 0.50 settings to align with aggressive retention goals.
- ▶ **High Coverage (85% Recall):** Successful capture 8.5 out of 10 actual churners for intervention.
- ▶ **Strategic Precision (47%):** Intentionally accepted a higher “false alarm” rate to ensure no high-value churners are missed in “Trouble Zone.”
- ▶ **Cost Justification:** The negligible cost of a signal check far outweighs the \$500+ loss of a subscriber.



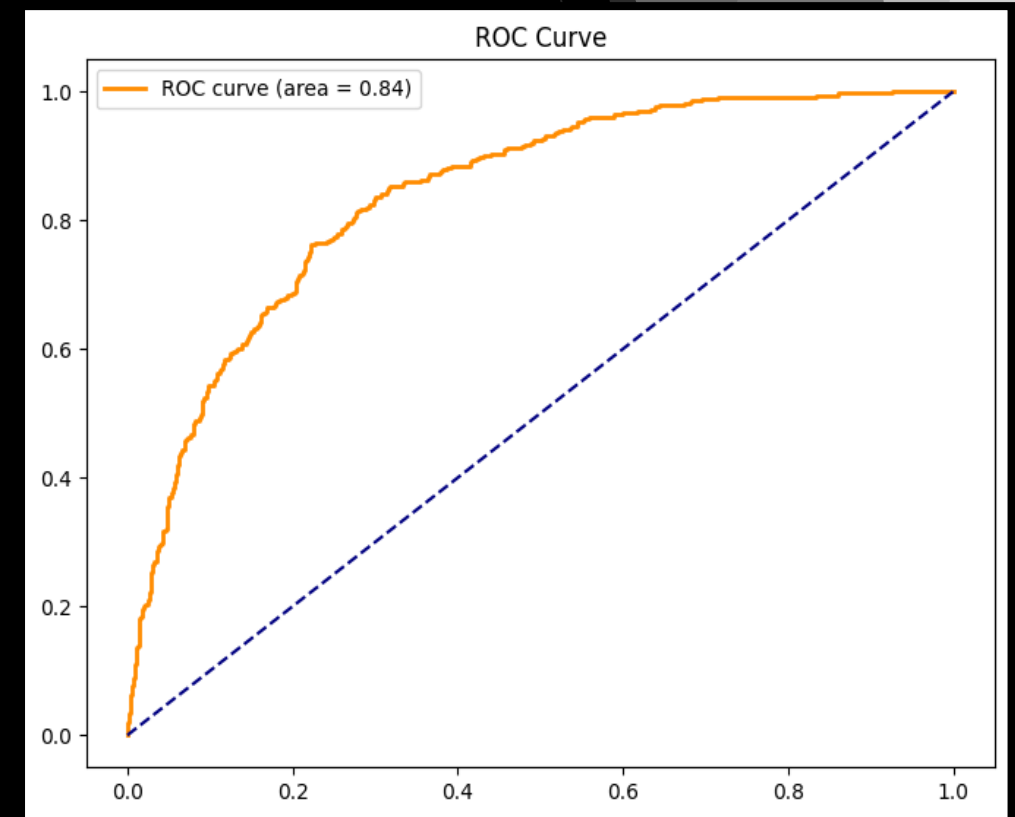
Power of Prediction (ROC Curve)

(Proving the model's 0.84 discriminative power)

- ▶ **Discriminative Power:**
 - ▶ The 0.84 AUC confirms the model effectively separates 'Stay' from 'Churn' well above the random baseline (0.50).
- ▶ **Early Detection Capability:**
 - ▶ The steep initial curve demonstrates high performance in identifying "obvious" churners.
 - ▶ Strongest performance is linked to customers in the 98104 Trouble Zone, where signal indicators are clear.

Takeaway:

Excellent discriminative power; the model successfully generalizes geospatial risk without overfitting.



Performance Diagnosis (Confusion Matrix)

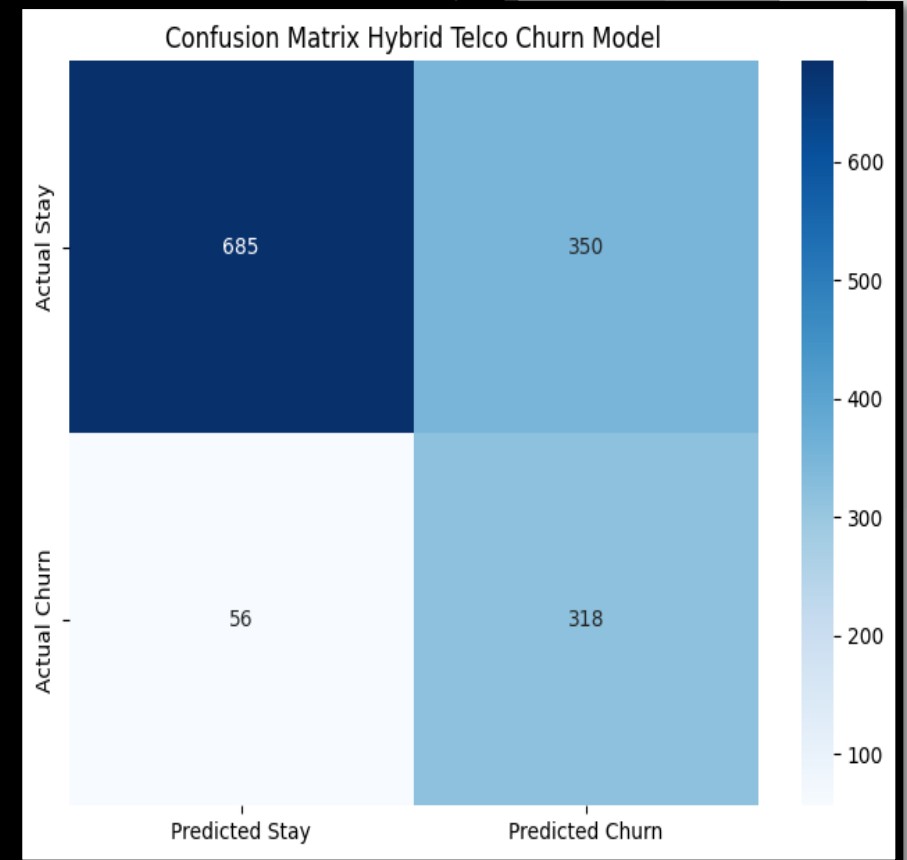
(The hard numbers: 318 Saves vs. 56 Misses)

Key Insights:

- ▶ **85% Churn Capture (318 True Positives):** The model successfully identified 318 out of 374 actual churners. This high recall ensures that "Trouble Zone" risk rarely goes unnoticed.
- ▶ **Minimal Blind Spots (Only 56 Misses):** Only 56 churners were missed by the system, proving the model is highly sensitive to the risk signals we engineered.
- ▶ **Strategic Trade-off (350 False Alarms):** Flagging these customers is an intentional business choice. A proactive "Signal Health Check" is negligible compared to the \$500+ loss of a Fiber subscriber.

Business Value:

- ▶ By shifting the threshold to 0.43, we have "tilted" the model to be over-inclusive. We would rather provide a proactive check-up to a loyal customer than miss a single high-value customer about to leave due to infrastructure failure.

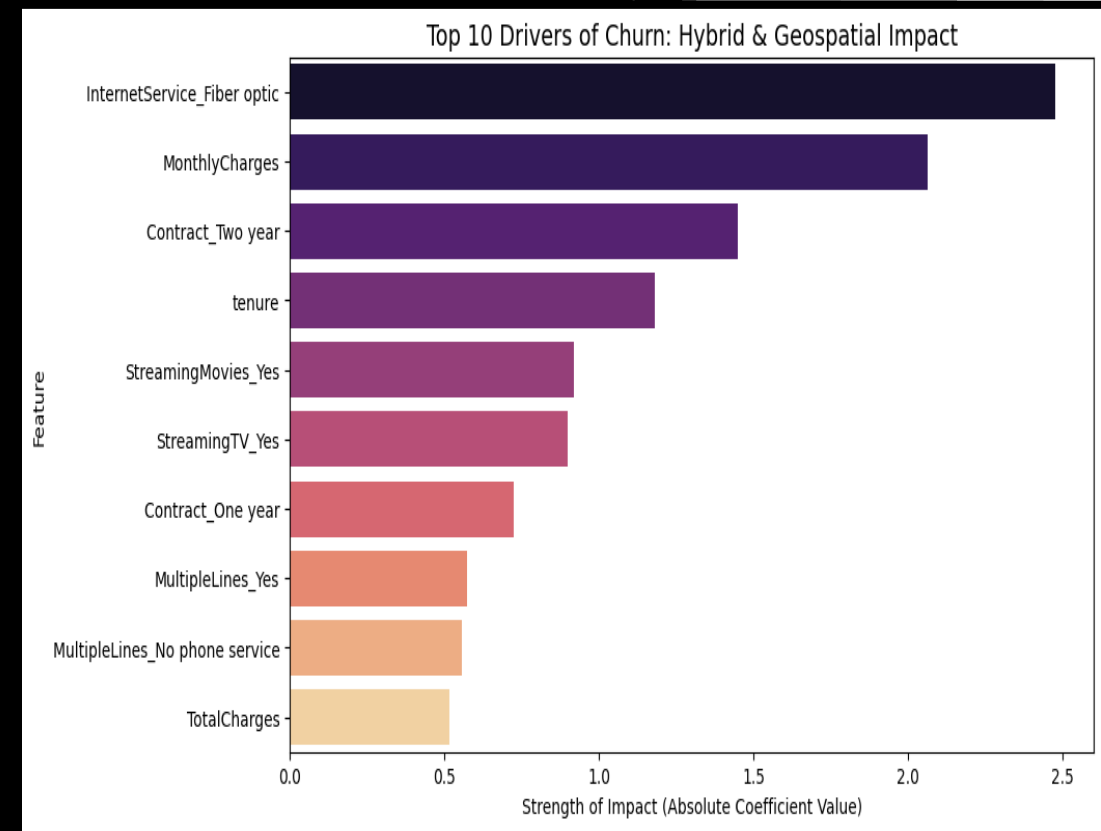


Identifying Key Churn Drivers

(The "Masking Effect"—Fiber Optic and Monthly Charges)

► Key Insights:

- **The #1 Risk (Fiber Optic):** Fiber customers are our highest-paying segment, but they have a 'zero-tolerance' policy for signal drops. This is where our revenue is most vulnerable.
- **Financial Anchors:** These anchors buy us time, but they won't stop a customer from leaving once their contract is up if the technical issues persist.
- **The "Masking" Effect:** Signal Stability isn't at the top of the chart because it's 'baked into' the Fiber and Streaming variables. If you fix the signal, you simultaneously stabilize our top three risk drivers.



Prescriptive Simulation (The ROI)

(Business Impact: The "What-If" Analysis)

The "What-If":

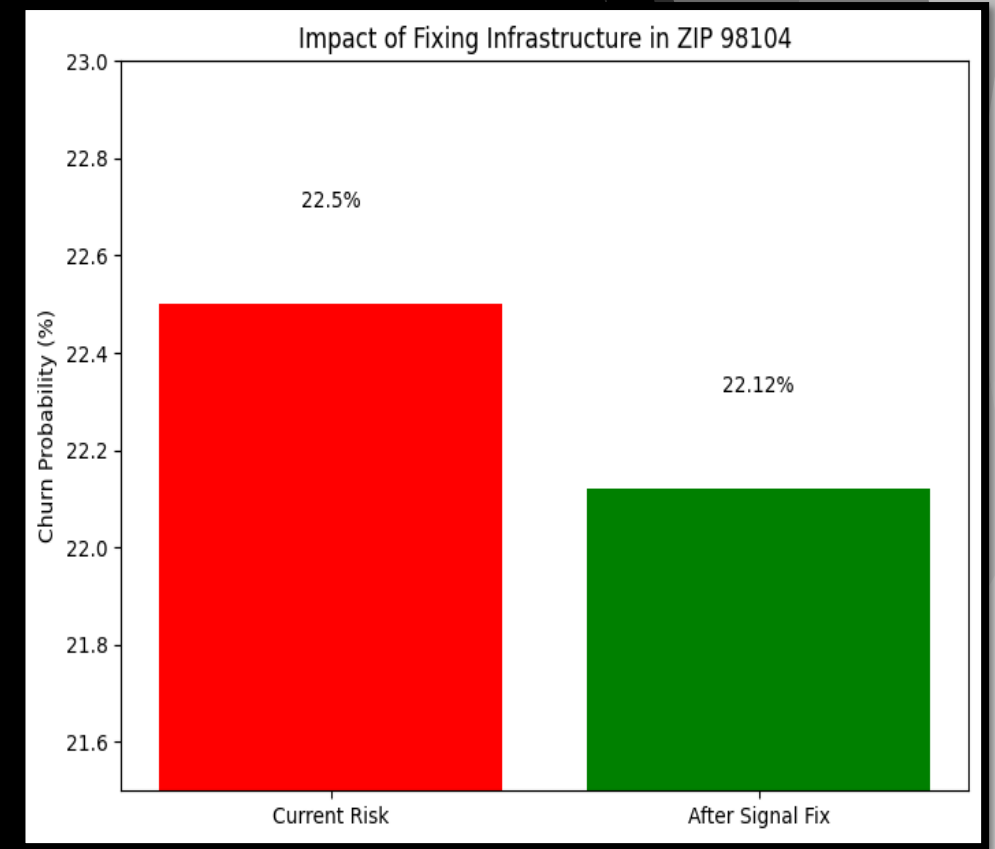
Improving signal in ZIP 98104 from 0.45 to 0.80.

The Financial Result:

- ▶ Pre-Fix Churn Risk: 22.50%
- ▶ Post-Fix Churn Risk: 22.12%

Impact:

The 1.69% relative reduction—equivalent to recovering 43 high-value accounts in a single zone.



Conclusion & Strategic Roadmap

▶ **Surgical Technical Intervention:**

- ▶ Fix Fiber-heavy ZIPs (98101-98105) when **Signal Stability** falls below 0.6.
- ▶ Capture the **1.69% relative churn reduction** without using “blanket” discounts.

▶ **Behavioral Anchoring (The Defensive Strategy):**

- ▶ Utilize **85% Recall** to identify at-risk **Fiber** users before they reach the “breaking point.”
- ▶ Incentivize **1-2 year contracts** to anchor behavior while infrastructure is stabilized.

▶ **Proactive Engagement (The Customer Experience):**

- ▶ Use **Outage History Scores** as a real-time trigger for localized outreach.
- ▶ Communication of technical fixes builds trust and lowers churn sensitivity.

To wrap up, our roadmap shifts the company from a reactive stance to a proactive, targeted approach. By prioritizing our Fiber-heavy 'Trouble Zones' the moment signal stability hits that 0.6 breaking point, we protect high-value revenue without relying on expensive, blanket discounts. By anchoring those same customers with behavioral incentives (contracts) while we stabilize the connection, we ensure that technical fixes translate into long-term loyalty. This hybrid approach doesn't just predict risk—it manages it at the root cause."

I end my presentation with this: *"Data-driven infrastructure leads to customer-driven loyalty."*

Thank you!

